Advanced Labs I - Theme E: Polymerization of Styrene

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1 Introduction

In this lab experiment, we focused on the synthesis of polystyrene which is a synthetic polymer used in different plastics, containers, etc. The experiment first synthesizes polystyrene using benzoyl peroxide as an initiator. Then we used the infrared absorption spectrum on the resulting polystyrene film and verified the results.

2 Methods and procedure

The production of of powdered polysterene and polystyrene film was as follows:

1. First, 4.9mg of the initiator benzoyl peroxide is measured using an automatic scale and added to the polymerization flask.
2. Using a micropippete we measured a 1mL of styrene to the polymerization flask.
3. Argon gas was injected into the flask for 30 seconds and then it was sealed using a stopper coated with silicon grease and held in place with rubber bands.
4. We shook the flask until the initiator was dissolved.
5. The flask was placed in a 75°C water bath for 24 hours.
6. The flask removed from the water bath, unsealed and a magnetic stirring bar was dropped in while the flask placed on the stirring machine.
7. 20mL of toluene was added to the vessel and the stirring machine was turned on until all of the styrene was dissolved.
8. We added 200 mL of ethanol into a flask and the flask was mounted onto the stirring machine. A magnetic bar was dropped.
9. The toluene solution was added into the ethanol and it was stirred until a cotton like precipitate appeared.
10. Using a suction filtration bottle, the solution was filtered to a obtain a solid precipitate of powdered polystyrene.
11. The powder was allowed to dry and later was transferred into a test tube and dissolved using a small amount of toluene.
12. The dissolved sample was poured into a beaker filled with water and left to dry. The result was a polystyrene film that could be used for testing.

The polysterene film was subjected to infrared absorbtion spectroscopy an infrared spectrophotometer to determine the bond types present.

3 Results

The powdered polystyrene was left at the lab and the film is attached to this report. The spectroscopy results are attached too.

4 Discussion

The results of the spectroscopy show multiple regions where the transmittance drops significantly namely 1, 2, 3, 4, 6, 7, 11, 12, and 13. Each of these regions correspond to wavelengths where molecular bonds absorb most energy. Peaks 1 to 4, located approximate to $3000^{cm^{-1}}$, correspond to the C-H molecular bond vibrations. This is present in the aromatic ring of the polystyrene molecule. The region around $1500^{cm^{-1}}$ has peaks no 6 and 7 which correspond to CH_2 and CH_3 bonds. The peaks 11 to 13 correspond to the rocking vibrations of CH_2 around $700^{cm^{-1}}$. These results are evidence that the resulting material is indeed polystyrene.